

RENANCE

PETROLEUM TRAINING INSTITUTE, EFFURUN

EEC 111 — Past Questions & Materials

by Renance

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Paper 1 of 1 — EEC 111 HANDOUT

Level ND1 • First Semester • Handout

WHAT IS SINGLE-LINE DIAGRAM?

A Single-Line Diagram, also known as SLD, is a type of electrical drawing which is used for representing an electrical system graphically. It is the most simplified visual representation of a complex electrical system.

In a single-line diagram of an electrical system, the main components of the system and their interconnections are depicted with the help of standardized symbols.

The main purpose of a single-line diagram or SLD is to provide a simplified drawing of configuration of a complex electrical system without showing any unnecessary details.

The single-line diagram helps to understand the architecture of an electrical system and provide ease in maintenance work.

Single-Line Diagram Example

The following figure depicts a typical single-line diagram of an electrical system.

ADVANTAGES OF SINGLE-LINE DIAGRAM

In the field of electrical engineering, the single-line diagrams (SLDs) are of great importance because of the following major advantages

- SLD is the most simplified graphical representation of a complex electrical system. Thus, it makes it easier for engineering professionals and technicians to understand the system architecture and interconnections of components.
- In a single-line diagram, not all the details about the system are mentioned. Hence, it is a compact representation of the system and easy to interpret.
- Since SLD is designed using standardized symbols of components. Thus, it is used as a common language for communicating system layouts among designers, engineers, and technicians.
- Preparing single-line diagrams is highly time efficient that enables designers and engineers to quickly visualize and interpret different system layouts and optimize them to comply with industry standards and regulations.
- A single-line diagram plays a crucial role in maintenance work as it helps engineers and technicians to determine system components and understand their interconnections which is very important in troubleshooting process. Further, the use of SLD greatly reduces the downtime of maintenance work.

DISADVANTAGES OF SINGLE-LINE DIAGRAM

Although, the single-line diagram (SLD) has several advantages as given above, but it also has some disadvantages or limitations, which are given below

- Single-line diagram does not provide detailed information about various components and interconnections of the system. Hence, it is not suitable to understand the complete

operation and behavior of an electrical system.

- Single-line diagram is a 2D drawing of an electrical system and it does not provide information about physical aspects like cable routing and component placement, etc.
- In a complex electrical system, which has some redundant components or multiple alternative paths for cables/wires, the single-line diagram becomes an inefficient visual representation.
- The complexity of single-line diagrams increases with the increase in size and complexity of the electrical system. Thus, for very large electrical systems, SLD is not an effective electrical drawing.

APPLICATIONS OF SINGLE-LINE DIAGRAM

The single-line diagram (SLD) is used in various applications in the field of electrical engineering. Some of the common applications of single-line diagrams are listed here

- Single-line diagrams are used during the initial design stage of electrical systems for planning and conceptualizing the configuration of system components like transformers, generators, distribution panels, circuit breakers, and other.
- Single-line diagrams are used for documentation of electrical installations.
- Single-line diagrams are also used for specifying the type, rating, and interconnections of electrical components required for a specific application.
- Electrical engineers and technicians use single-line diagrams to accomplish installation work of electrical systems.
- During the testing and commissioning, single-line diagrams are used to verify that all electrical components of the system have been installed correctly.
- During the testing and commissioning, single-line diagrams are used to verify that all electrical components of the system have been installed correctly.
- Single-line diagrams are also helpful in upgrading or modifying the existing electrical system.

WHAT IS MULTILINE DIAGRAM?

In electrical design drawing, a multiline diagram is a graphical tool used for visualizing three-phase electrical power systems or power delivery networks. Thus, it is also known as three-line diagram. In a typical multiline diagram, we use multiple parallel lines to depict the different phases and circuit components of the electrical system.

Similar to single-line diagrams, the multiline diagrams are also used in electrical design and documentation applications. Using multiline diagrams, electrical engineers and technicians can obtain more detailed and accurate information about the electrical infrastructure.

Multiline Diagram Example

An example of multiline diagram of an electrical system is shown in the following figure.

ADVANTAGES OF MULTILINE DIAGRAM

Multiline diagrams are extensively used for representing three-phase electrical systems

graphically because of the following major advantages

- Multiline diagram allows us to represent each phase of a three-phase electrical system separately by using parallel lines. This gives a clear and detailed visualization of a multiphase electrical system.
- In distribution networks, multiline diagrams provide a way of illustrating busbars and conductors differently. Where, busbars are visualized as thick lines, while conductors are represented as thin lines.
- Multiline diagrams provide a detailed view of interconnections between circuit components like transformers, generators, circuit breakers, and electrical loads.
- Multiline diagrams are effective in representing large and complex electrical systems.

DISADVANTAGES OF MULTILINE DIAGRAM

However, multiline diagrams offer several advantages as mentioned above, but they also have some disadvantages/limitations over single-line diagrams as listed below

- Multiline diagrams are relatively more complex as compared to single-line diagrams. This is because they use separate representation of each phase using parallel lines.
- Multiline diagrams are relatively harder to understand and interpret for persons who are not having knowledge of electrical engineering.
- Multiline diagrams use more lines and symbols; hence they require more space than single-line diagrams to represent electrical systems.
- Sometimes, multiline diagrams can be confusing due to use of multiple lines. It happens more when lines intersect or overlap. Thus, a clear and proper labeling and organization of lines and components is required.
- Multiline diagrams involve more complex documentation of electrical systems.
- Multiline diagrams are less consistent and standardized across different industries, areas, and organizations.
- For multiline diagrams, it is relatively difficult to modify and update them and they require more attention while revising.
- Multiline diagrams are more prone to errors and inconsistencies.

APPLICATIONS OF MULTILINE DIAGRAM

In the field of electrical engineering, multiline diagrams are used in the design and documentation of complex electrical systems. Some examples of applications of multiline diagrams are listed below

- Multiline diagrams are extensively used in electrical designs of power systems, substations, distribution systems, and other industrial electrical systems.
- Multiline diagrams are also used for graphical representation of three-phase electrical systems.
- Multiline diagrams also play an important role in load flow analysis in power systems, as they allow engineers to model the flow of current and power between nodes and analyze

the performance of the systems to meet the specific requirements.

- Multiline diagrams act as detailed document of an electrical system and provide complete information about various electrical installations within the system.

CONCLUSION

In electrical engineering, both single-line diagram (SLD) and multiline diagram (MLD) are essential tools used for design and documentation purposes of electrical systems.

A single-line diagram is a most simplified graphical representation of electrical systems in which standard symbols and lines are used to represent interconnections of the system. SLDs are mainly used for initial design and basic documentation purposes, and they are not suitable for representation of complex systems accurately.

On the other side, a multiline diagram is a more detailed representation of an electrical system and provides all the important information about the system architecture, phase relationship, and more. Multiline diagrams are powerful design tools for representation and documentation of large and complex electrical systems like power distribution networks.

DIFFERENCES BETWEEN SYMBOLIC AND COMPONENT DIAGRAMS

In electrical and electronics engineering, circuits can be represented in different ways depending on the purpose. The two common types are:

- Symbolic Diagram
- Component Diagram

1. Symbolic Diagram (Schematic Diagram)

A symbolic diagram uses standardized electrical symbols to represent components and show how they are electrically connected.

Features:

- Uses international standard symbols (IEC/IEEE).
- Shows electrical connections clearly.
- Does NOT show the physical appearance or actual size.
- Focuses on how the circuit works.

Example:

A resistor is shown as a zigzag line.

A battery is shown as long and short parallel lines.

A transistor is shown with three labeled terminals.

Purpose:

- Circuit design and analysis.
- Troubleshooting.
- Understanding circuit operation.

2. Component Diagram (Pictorial / Physical Diagram)

A component diagram shows the actual physical appearance of components as they look in real

life.

Features:

- Shows real shapes and sizes.
- May include color coding.
- Shows approximate physical layout.
- Easier for beginners to recognize parts.

Example:

- A real picture of a resistor with color bands.
- A real battery with terminals marked (+) and (–).
- A real switch with its lever.

Purpose:

- Installation and assembly.
- Training beginners.
- Identifying components physically.

Key Differences

Feature	Symbolic Diagram	Component Diagram
Representation	Uses standard symbols	Uses real-life pictures
Focus	Electrical function	Physical appearance
Used For	Design and analysis	Installation and identification
Technical Level	More technical	Easier for beginners
Shows Size/Shape	No	Yes

How to Use CAD in Creating Electrical and Electronic Drawings

The general steps for using CAD software are similar across most programs:

1. Start a New Project

- Open the software
- Select drawing type (electrical schematic, wiring, PCB, etc.)
- Set drawing units and standards

2. Select Symbols or Components

- Choose components from the software library such as:

- o Resistors

- o Switches

- o Motors

- o Transformers

- o Relays

- o ICs

Most CAD software provides standard symbol libraries.

3. Place Components on the Drawing Sheet

- Drag and drop symbols into the workspace

- Arrange them logically according to circuit flow
4. Connect Components
- Use wiring tools to connect components with lines
 - Add junction points where necessary

5. Add Labels and Codes

- Component names (R1, C1, M1)
- Wire numbers
- Voltage ratings
- Terminal numbers

6. Verify the Design

- Check for connection errors
- Confirm ratings and specifications
- Run simulation (if software supports it)

7. Save and Print

- Export drawing as PDF or image
- Print for installation or documentation

COMMON SOFTWARE USED FOR ELECTRICAL AND ELECTRONIC DRAWINGS

Below are some widely used CAD software tools:

1. AutoCAD Electrical

Developed by Autodesk.

Features:

- Professional electrical drawing tools
- Large symbol libraries
- Automatic wire numbering
- Panel layout design
- PLC design support

Used for:

- Industrial control systems
- Building electrical installations
- Power distribution drawings

2. EPLAN

Developed by EPLAN GmbH & Co. KG.

Features:

- Advanced electrical engineering design
- Automation system design
- Control panel documentation
- Automatic reports and bill of materials

Used for:

- Industrial automation
- Manufacturing plants
- Complex electrical systems

3. PC Schematic

Developed by PC SCHEMATIC A/S.

Features:

- User-friendly interface
- Electrical documentation tools
- Cable and terminal diagrams
- Automatic numbering

Used for:

- Electrical control systems
- Machine wiring diagrams
- Industrial documentation

4. CircuitMaker 2000

Developed by Altium Limited.

Features:

- Electronic circuit design
- Simulation capabilities
- PCB design tools
- Component libraries

Used for:

- Electronic circuit design
- PCB development
- Educational purposes
- List of Electrical Components With Symbols
- An electrical symbol is a graphical representation of electrical and electronic components. These symbols help us recognise a particular electronic device in a circuit. National and international standards define electrical symbols. Circuit symbols only represent the components of electrical and electronic circuits and do not define their function or process.
- Let us have a look at how various components in a circuit are denoted:
- Electrical Wire
- A wire is a single, usually cylindrical, flexible strand or rod of metal through which electric current flows. It is usually made of good conducting metals such as copper.
- Connected Wire
- Connecting wires provide a medium to an electrical current so that they can travel from one point on a circuit to another.
- Disconnected Wire
- In diagrams, we come across situations where we have wires crossing through other wires even though they are not connected to each other.

Hence, it is preferred to have a hump as shown to depict the crossing of one wire over another wire.

- Earth Ground
- The reference point in an electrical circuit from which voltages are measured, a common return path for electric current, or a direct physical connection to the earth.
- Chassis Ground
- Chassis ground is the metal housing that some electrical device is encased in. The chassis may be connected to the green ground wire of your power lines.
- Digital/Common Ground
- The reference voltage of digital logic ICs. This means that no analogue signals are coupled into this reference plane.
- Lamp/Light Bulb
- It is a load that uses electric current to emanate light.
- Resistor (IEE)/(IEC)
- A resistor is a device that resists the flow of current through a circuit.
- Variable Resistor/ Rheostat (IEE)/(IEC)
- A variable resistor is a resistor whose electrical resistance can be adjusted. A rheostat is a variable resistor whose resistance can be varied to control the current in the circuit.
- Potentiometer (IEE)/(IEC)
- A potentiometer is a three-terminal resistor with a sliding or rotating contact that forms an adjustable voltage divider.
- Photoresistor/ Light-dependent resistor (LDR)
- Photoresistor decreases resistance with respect to receiving luminosity.
- Thermistor
- A thermistor is a type of resistor whose resistance is dependent on temperature.
- Capacitor
- A capacitor is a device that stores electrical energy in an electric field.
- Variable Capacitor
- A variable capacitor is a capacitor whose capacitance may be intentionally and repeatedly changed mechanically or electronically.
- Polarized Capacitor
- A capacitor whose anode is made of metal forms an insulating oxide layer through anodisation.
- Voltage Source/ Current Source
- A voltage source is an electrical component that can maintain a fixed

voltage irrespective of the load resistance and output current.

- Battery Cell/ Battery
- A battery is a device that consists of two or more electrochemical cells with external connections.
- AC Voltage Source
- It is a source whose positive and negative terminals change periodically.
- Controlled Voltage Source/ Controlled Current Source
- A current-controlled voltage source produces a voltage that is linearly proportional to the current. A controlled or dependent current source, on the other hand, changes its available current depending upon the voltage.
- Voltmeter
- The voltmeter is an instrument for measuring electrical potential.
- Ammeter
- The Ammeter is an instrument for measuring electric current.
- Diode
- The diode is a semiconductor device with two terminals that allows current to flow only in one direction.
- Zener Diode
- The Zener diode is a type of diode that allows the flow of reverse current when it reaches a certain Zener voltage.
- Schottky Diode
- The Schottky Diode is a diode formed by the junction of a semiconductor with a metal.
- Varicap Diode
- The Varicap diode is a type of diode designed to exploit the voltage-dependent capacitance of a reverse-biased p-n junction.
- Photodiode
- The photodiode is a type of semiconductor device that converts light into an electrical current.
- NPN Bipolar Transistor
- The NPN bipolar transistor is the transistor in which one p-type material is placed between two n-type materials.
- PNP Bipolar Transistor
- PNP bipolar transistor is the transistor in which one n-type material is placed between two p-type materials.
- OR/NOR Gate
- OR gate performs a logical operation which gives the value one if at least one operand has the value one and otherwise gives a value of

zero.

NOR gives the value one if and only if all operands have a value of zero and otherwise has a value of zero.

- **AND/NAND Gate**

- AND gate gives the value one if and only if all the operands are one, and otherwise has a value of zero.

NAND gate gives the value zero if and only if all the operands have a value of one, and otherwise have a value of one.

- **XOR/ NOT Gate**

- XOR gate is a digital logic gate that gives a true output when the number of true inputs is odd.

NOT gate is a logic gate which implements logical negation.

Simple electrical installation diagrams use standardized, abstract symbols connected by lines to represent physical wiring, components (switches, lights, outlets), and their layout in a circuit.

Key basics include dots for connected wires, gaps for unconnected, and standardized shapes for devices to safely guide installation, following electrical codes.

- **Lines:** Represent wires/conductors. A solid dot where lines cross indicates a connection; a "bridge" or jump means they cross without connecting.
- **Power Source:** Usually shown as a circle with "AC" (wave symbol) or for DC. +/-
- **Switch:** A break in the line, appearing as a lever, that opens or closes a circuit.
- **Light/Load:** A circle with an "X" inside, or a circle with a tail (wall scone).
- **Socket/Outlet:** Typically, a circle with two lines extending from it, or two parallel lines.
- **Ground:** A vertical line connected to three decreasing horizontal lines.
- **Circuit Breaker/Fuse:** A zigzag line or a box with a line through it, indicating protection.

Example: Simple Light Switch Diagram

1. **Line (Hot/Live):** Connects from the breaker box (L) to the switch.
2. **Switch:** The switch connects/disconnects the hot wire.
3. **Load:** Another line travels from the switch to the light fixture (X).
4. **Neutral:** A separate neutral wire connects directly from the power source to the light fixture to complete the circuit (N).
5. **Ground:** A green or bare wire connects the metal casing of the fixture to the ground point.

Tips for Reading Diagrams

- **Start at the source:** Locate the power source or distribution panel.
- **Follow the path:** Trace the path of current from hot (L) to load and back to neutral (N).
- **Check connections:** Ensure you distinguish between connected (dotted) and crossing (unconnected) wires.
- **Lettering:** Use standard letter codes (e.g., L1, L2, N) to identify

HOW TO READ ELECTRICAL AND ELECTRONIC DRAWINGS

Reading electrical and electronic drawings means understanding the symbols, connections, and information shown in a circuit diagram so you can know how a system works, install it correctly, or troubleshoot faults.

Below is a step-by-step guide you can follow:

1. Read the Title Block First

The title block contains important information such as:

- Name of the project or circuit
- Drawing number
- Date
- Author or company
- Revision number
- Scale (if applicable)

2. Identify the Type of Drawing

Determine what kind of diagram you are looking at:

- Single line diagram → Overview of power system
- Schematic (symbolic) diagram → Circuit operation
- Wiring diagram → Physical connections
- Layout diagram → Component positions

3. Understand the Symbols Used

Learn the standard symbols for components such as:

- Resistors
- Switches
- Transformers
- Motors
- Diodes and transistors
- Relays and fuses

4. Follow the Circuit Path

Trace the circuit from:

Power source → Control devices → Load → Return path

Example:

Power → Switch → Fuse → Lamp → Neutral

This helps you understand how current flows.

5. Check Labels and Codes

Electrical drawings include identification codes such as:

- L, N, E (Live, Neutral, Earth)
- R, Y, B (Three phases)
- Terminal numbers (1, 2, 3...)
- Component tags (M1 = Motor 1, CB1 = Circuit Breaker 1)

6. Observe Connection Points

Look for:

- Junction dots (connected wires)

- Crossing lines without dots (not connected)
- Terminal blocks
- Connector pins

7. Understand Ratings and Values

Components often include specifications such as:

- Voltage rating (230 V, 415 V)
- Resistance value (10 kΩ)
- Current rating (5 A)
- Power rating (100 W)

8. Look for Notes and Legends

The legend explains symbols used in the drawing.

Notes may include:

- Installation instructions
- Safety warnings
- Special conditions

9. Break Complex Circuits into Sections

Large drawings can be confusing.

Divide them into parts:

- Power section
- Control section
- Protection section
- Output/load section

10. Practice Visualization

Try to imagine:

- How current flows
- Which device controls another
- What happens when a switch is ON or OFF

DIFFERENCE BETWEEN SINGLE LINE AND MULTI LINE

- Feature • Single Line • Multi Line
- Lines used • One line represents all phases • Each wire drawn separately
- Detail level • Low • High
- Usage • Planning, analysis • Installation, maintenance
- Complexity • Simple • Detailed

Types of Electrical and Electronics Symbols

Electrical and electronic symbols are graphical representations used in circuit diagrams to show components and their connections clearly without writing long descriptions. They are grouped based on the function of the components they represent.

1. Power / Electrical Symbols

Used in electrical power systems and installations.

Examples:

- Generators and motors
- Transformers
- Switchgear (circuit breakers, isolators)
- Transmission lines
- Earthing (ground) systems

Purpose: Used in power distribution, industrial wiring, and electrical installations.

2. Electronic Symbols

Used in electronic circuits for signal processing and control.

Examples:

- Semiconductor devices (diodes, transistors, SCRs)
- Integrated circuits (ICs)
- Resistors, capacitors, inductors
- Logic gates (AND, OR, NOT)

Purpose: Used in communication systems, computers, automation, and electronic devices.

3. Control and Instrumentation Symbols

Used in automation and measurement systems.

Examples:

- Sensors
- Relays and contactors
- Control switches
- Meters (voltmeter, ammeter, wattmeter)

Purpose: Used in industrial control panels and monitoring systems.

4. Wiring and Installation Symbols

Used in building wiring diagrams.

Examples:

- Lamps and lighting points
- Socket outlets
- Distribution boards
- Switches

Purpose: Used by electricians during installation and maintenance.

DIFFERENT STANDARDS OF SYMBOLS

Electrical symbols are standardized worldwide to ensure engineers from different countries can understand diagrams easily. The main standards organizations include:

1. International Electrotechnical Commission (IEC)

- International organization that prepares and publishes standards for electrical and electronic technologies.
- IEC symbols are widely used in Europe, Africa, and many parts of the world.
- Provides uniform graphical symbols under IEC 60617 standard.

2. Institute of Electrical and Electronics Engineers (IEEE)

- A global professional organization for electrical and electronics engineering.
- Develops technical standards mainly for electronics, communications, and computing.

- IEEE symbols are common in academic and research environments.

3. American National Standards Institute (ANSI)

- Coordinates standards used in the United States.
- ANSI electrical symbols are widely used in North America.
- Many industrial drawings and schematics in the U.S. follow ANSI standards.

4. Japanese Electrotechnical Committee (JEC)

- Responsible for electrical standards in Japan.
- Works closely with IEC but includes national adaptations.
- Used mainly in Japanese electrical equipment and documentation.